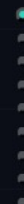


GLOBAL ENERGY REPORT · 2025

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix – spanning hydropower, wind, and solar technologies.

Data: Our World in Data · Ember (2026) · Energy Institute Statistical Review (2025)



75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse gas emissions
from fossil fuel combustion

5M+

premature deaths annually from
fossil fuel air pollution

Renewables Now Supply ~14% of Global Primary Energy

~14% OF PRIMARY ENERGY

Roughly one-seventh of total global primary energy (electricity, transport & heating combined) came from renewables as of 2024 — measured using the substitution method.

WHY LOWER THAN ELECTRICITY?

Transport and heating remain far more reliant on oil and gas — pulling the overall share well below the electricity-sector figure of ~30%.

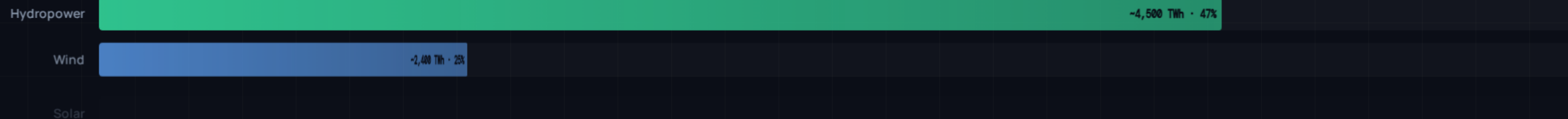
STRONG REGIONAL VARIATION

Norway, Brazil, and New Zealand derive large shares from renewables; fossil-fuel producers in the Middle East and parts of Asia remain below 5%.



Renewable Electricity Generation Is Growing Rapidly, Led by **Wind** & **Solar**

■ Hydropower ■ Wind ■ Solar ■ Other renewables



Almost **One-Third** of Global Electricity Now Comes from Renewables



COUNTRY / REGION	RENEW. SHARE	
Norway	~98%	
Brazil	~85%	
Canada	~65%	
China	~35%	
India	~25%	
Saudi Arabia	<2%	
World avg.	~30%	

Hydropower Remains the Largest Renewable Source at ~50% of Renewable Generation

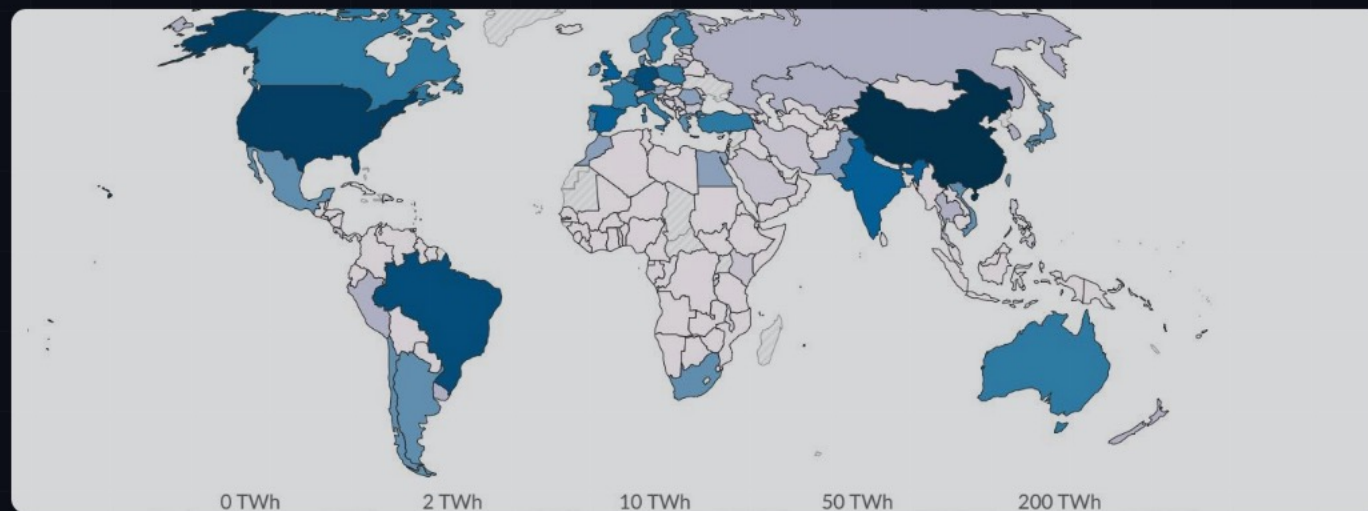
~4,500

TWh generated globally in 2024 — roughly half of all renewable electricity

Hydropower has been generating electricity at scale for over a century. Its growth potential is now more constrained than wind and solar due to geography and environmental factors.

TOP PRODUCERS

#1	China	World's largest hydro producer
#2	Brazil	~85% of electricity from renew.
#3	Canada	~65% of electricity from renew.



Wind & Solar Are the **Fastest-Growing** Energy Technologies in History

Wind Energy

~2,400 TWh

25% of renewable electricity (2024)

Major contributor across Europe, China, and the United States. Includes onshore and offshore installations.

Solar Energy

~2,000 TWh

21% of renewable electricity (2024)

Fastest-growing source globally — driven by dramatically falling costs and supportive policies since 2010.

Wind + Solar combined (~4,400 TWh) now nearly match hydropower's ~4,500 TWh — and are growing far faster, making them the principal drivers of the renewable transition.



Stark Regional Gaps: Scandinavia & Latin America Lead While Sub-Saharan Africa & Middle East Lag

LEADERS (60-98%)

Norway (~98%), Brazil (~85%), Canada (~65%): abundant hydropower resources have enabled very high renewable electricity shares.

WIND & SOLAR CHAMPIONS

Denmark and Germany (~55%+): heavy investment in wind and solar has driven rapid decarbonisation of their electricity grids.

MIDDLE EAST (<10%)

Saudi Arabia (<2%), most Gulf states: remain almost entirely dependent on fossil fuels for electricity generation.

SUB-SAHARAN AFRICA

Minimal renewable electricity infrastructure despite abundant solar potential — South Africa ~10%, most nations far below.



Key Takeaways: Renewables Are Growing Fast but the **Transition Must Accelerate**

01

~14% of primary energy, ~30% of electricity

Renewables have reached significant milestones, but remain well short of the levels needed to meet climate targets. Transport and heating lag far behind the electricity sector.

02

Hydropower still dominant — but wind & solar are catching up fast

Hydro contributes ~47% of renewable electricity (~4,500 TWh) but its growth potential is limited. Wind (~2,400 TWh) and solar (~2,000 TWh) combined now nearly match hydro's output.